Name of Experiment: Histogram Analysis and Thresholding of a Grayscale Image.

Introduction: In this experiment, a grayscale image was loaded using **OpenCV** and its histogram was analyzed to observe the distribution of pixel intensities. Then, thresholding was applied to convert the grayscale image into a black-and-white binary image using a fixed threshold value. Histograms of both the original and binary images were displayed using Matplotlib to compare the changes in pixel distribution.

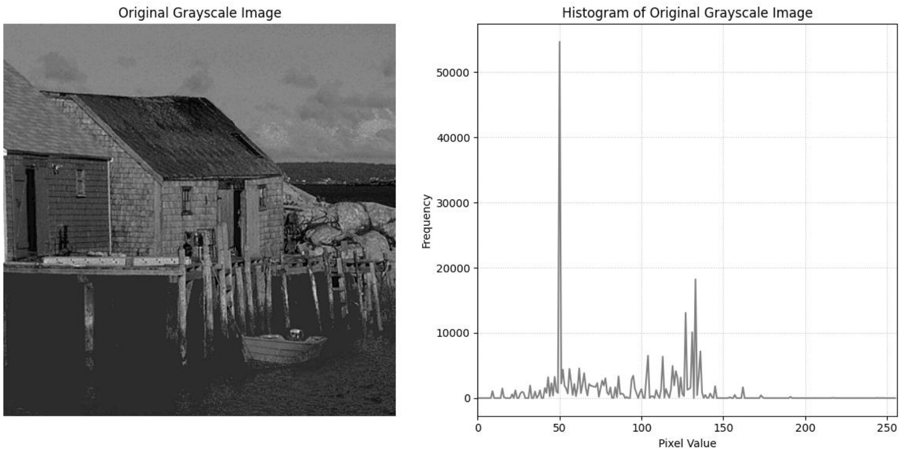
Code:

```python
import cv2
import numpy as np
import matplotlib.pyplot as plt
import os

image_path = "grayscale.gif"

if not os.path.exists(image_path):
    raise FileNotFoundError(f"Error: The file '{image_path}' does not exist.")

img_gray = cv2.imread(image_path, cv2.IMREAD_GRAYSCALE)

if img_gray is None:
    raise ValueError(f"Error: Could not load image from {image_path}")

print(f"Image '{image_path}' loaded successfully.")
print(f"Image Shape: {img_gray.shape}")


hist_gray = cv2.calcHist([img_gray], [0], None, [256], [0, 256])

plt.figure(figsize=(12, 6))

plt.subplot(1, 2, 1)
plt.imshow(img_gray, cmap='gray')
plt.title('Original Grayscale Image')
plt.axis('off')

plt.subplot(1, 2, 2)
plt.plot(hist_gray.ravel(), color='gray')
plt.title('Histogram of Original Grayscale Image')
plt.xlabel('Pixel Value')
plt.ylabel('Frequency')
plt.xlim([0, 256])
plt.grid(True, linestyle=':', alpha=0.7)

plt.tight_layout()
plt.savefig('output_1.png')
plt.show()

threshold_value = 127
_, img_binary = cv2.threshold(
    img_gray,
    threshold_value,
    255,
    cv2.THRESH_BINARY
)

hist_binary = cv2.calcHist([img_binary], [0], None, [256], [0, 256])

plt.figure(figsize=(12, 6))

plt.subplot(1, 2, 1)
plt.imshow(img_binary, cmap='gray')
plt.title(f'Black and White Image (Threshold = {threshold_value})')
plt.axis('off')
plt.subplot(1, 2, 2)
```

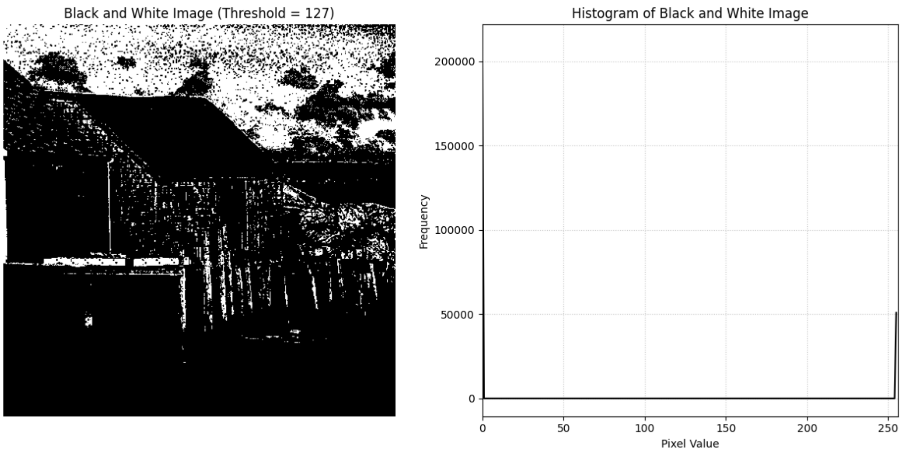
```
plt.plot(hist_binary.ravel(), color='black')
plt.title('Histogram of Black and White Image')
plt.xlabel('Pixel Value')
plt.ylabel('Frequency')
plt.xlim([0, 256])
plt.grid(True, linestyle=':', alpha=0.7)

plt.tight_layout()
plt.savefig('output_2.png')
plt.show()
```

<u>Input Image</u>:

<u>Output Image-1</u>:



<u>Output Image-2</u>:



<u>Discussion</u>: The grayscale image histogram shows the distribution of different gray intensity values in the image. After applying thresholding, the image is converted into only black and white pixels based on the threshold value. The binary image histogram mainly contains two intensity values, 0 and 255, representing black and white regions. This experiment demonstrates how thresholding simplifies an image and highlights important regions for image processing tasks.